

NATIONAL INSTITUTE OF BUSINESS MANAGEMENT
HIGHER NATIONAL DIPLOMA IN SOFTWARE ENGINEERING



ROBOTICS APPLICATION DEVELOPMENT

Final Project Report

Water Hyacinth Remover Robot (AquaHyel)

SUBMITTED BY

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Abstract

Water Hyacinth (*Eichhornia crassipes*) is an aquatic plant that was introduced to Sri Lanka in 1905. At first, it was considered as an ornamental plant then it turned to be an invasive plant that grows and spread rapidly while harming water bodies and its eco system. This project presents as a solution to remove the plant from water bodies while it is being portable, cost-effective and easy to implement with minimum requirements. 'AquaHyel' robot designed to achieve above mentioned standards powered by Arduino Uno, dc motors and conveyor belt as the main hyacinth remover mechanism. Proposed system minimizes the initiating of manual labour for the removal process and provide a time saving and efficient way heel the water bodies.

Key Words: Water Hyacinth, Water Hyacinth Remover, River Waste Cleaner, Water Body Maintenance, River Cleaner, Robotics Project, Arduino Uno

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Water Hyacinth Remover Robot (AquaHyel)

1. Introduction

Water hyacinth rapidly grows on water bodies (rivers, streams, canals, ponds etc.) as an invasive plant completely hindering the sunlight penetration causing reduced oxygen level underwater harming the overall ecosystem of the water bodies. It creates dense mat over the water surface (parallel to the water level).



Figure 1

Traditional approaches have been widely used for the water hyacinth removal process but many of them request manual labour. It can be costly, not efficient and not reliable since removal process needed to be repetitive and consistence attention for a longer period. ‘AquaHyel’ is fully automated robot which utilize minimum power and work with reliable components, with portable and removable structure which Is cost effective and flexible for the future enhancements.

2. Methodology

2.1 Problem definition

Analysing the biological features and the spreading behaviour of the to form effective removal mechanism.

2.2 Review current solutions

Review of the manual and automated systems which have been initiated and implemented supported to the identify the current limitations and challenges that those inventors have faced. It depicted the gaps in in exciting solutions and ways to enhance and improve our solution with better functionalities.

2.3 Developing a model

Architectural diagram is designed to indicate the identified functionalities required by the robot along with the power supply unit. Then required components were selected to develop the model. Then the circuit diagram is developed. Following that 3D design is designed to create the structure of the prototype. It helps to articulate the basic structure of the robot while identifying the possible issues and challenges that we must have to face. Further, we developed a basic prototype prior to the actual one including the recognised features using a temporary structure. Then the final prototype was developed.

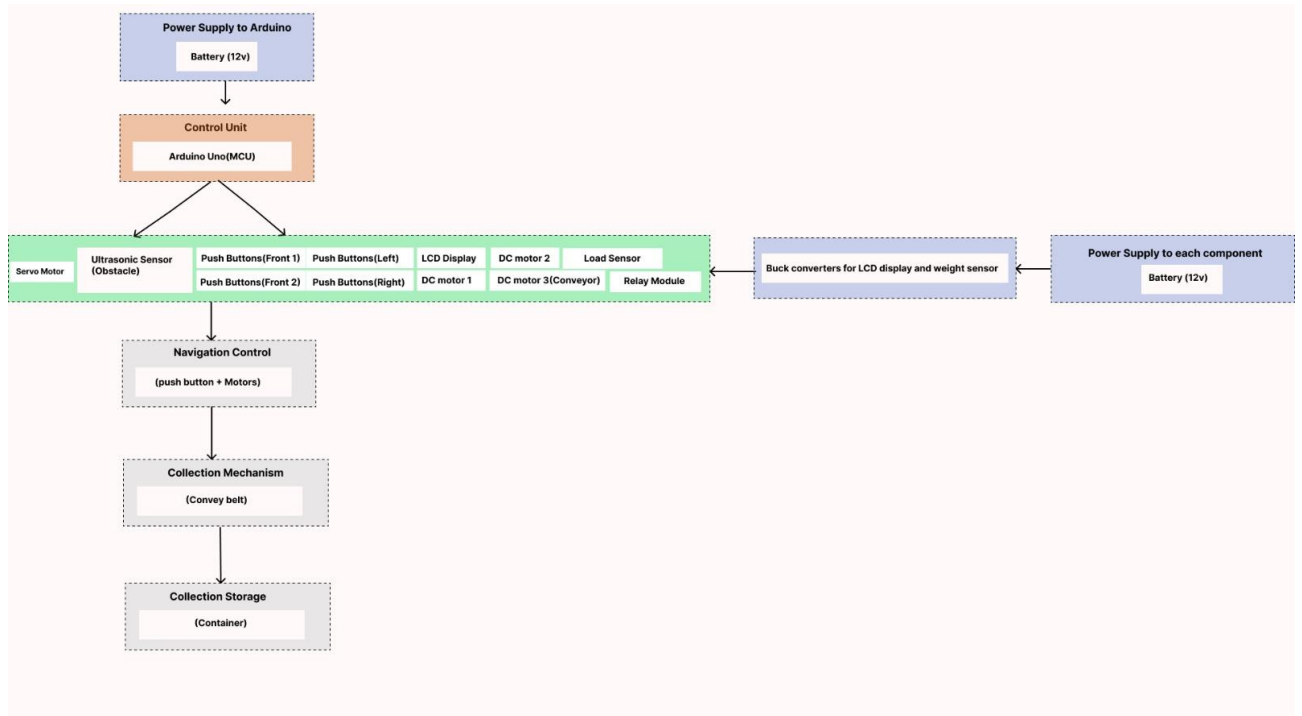


Figure 2

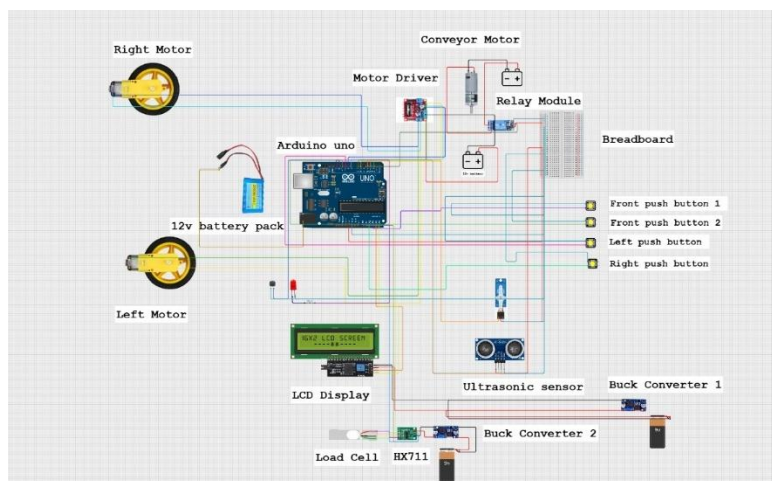


Figure 3

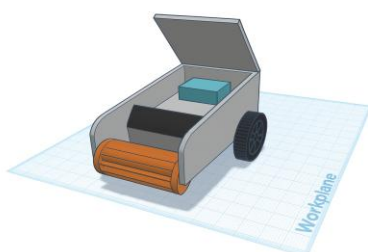


Figure 4

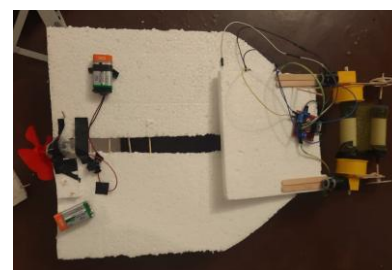


Figure 5

2.4 Designing the prototype

2.4.1 Structure and physical configuration

Base structure is created using the PVC pipes to make it strong lite and firm. Structure provide support to float on the water surface while it contains 3 main compartments.

- Fore compartment – Conveyer Belt to collect plants and push buttons to indicate edges.
- Middle compartment – Load cell to calibrate the weight and the bin to collect plants.
- Rear Compartment – Control unit and gear motors to move the boat.

Base structure is supported by the polystyrene (EPS) for the floating purposes and the middle compartment is supported by a plastic container.

2.4.2 Components review

I. Arduino UNO



Figure 6

- To connect components and provide signals accordingly.
- Powered by 3 (3.7v) lithium-ion batteries to get 12v (approximately) output.

II. Gear Motor 12V 200RPM XC25GA370



Figure 7

- To rotate the conveyer belt.
- Powered by 2 (3.7v) lithium-ion batteries to get 7.4v (approximately) output.

III. 5VDC 1 Way 1 Channel Relay Module (Transistor Version) (MD0486)



Figure 8

- Control power to the conveyer belt.
- Supply regulated power by microcontroller.

IV. SG90 Servo Motor



- Control the lifting bucket in the front to support the collection.
- Supply regulated power by microcontroller.

Figure 9

V. L298N motor driver



- Control the back-and-forth movement of the motor.
- Powered by batteries 9V(Approximately).

Figure 10

VI. Buck Converters



- Regulate power supply to the components.
- Supply regulated 5v to the LCD and the load cell.

Figure 11

VII. Gear motors



- To move the robot forward, backward, left and right movement.
- Powered by L298N motor driver.

Figure 12

VIII. HY-SRF05 HC-SR05 5Pin Ultrasonic Sensor Module (MD0080)



- Detect water hyacinth plants on the water surface.
- Supply regulated power by microcontroller.

Figure 13

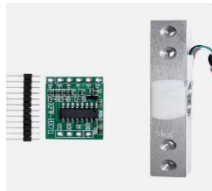
IX. Push buttons



- To indicate the edges of the water body (front two buttons and buttons in left and right sides).
- Supply regulated power by microcontroller.

Figure 14

X. Load Cell HX711 Module Weight Sensor 5kg Aluminium Alloy



- To measure the weight of the plants.
- Powered by 9v battery and power down to 5v using buck converters.

Figure 15

1602 LCD Display



- To display all messages to the user.
- Powered by 9v battery and power down to 5v using buck converters.

Figure 16

XI. Bulb & Buzzer



- To indicate that robot has reached its maximum weight and in the returning status.
- Supply regulated power by microcontroller.

Figure 17

XII. Breadboard



Figure 18

2.5 Testing and Implementation

2.5.1 Unit testing

Components tested individually to check the functionality of them.

2.5.2 Integration testing

All components were tested to check how they work together as a unit.



Figure 19

2.5.3 Functional testing

Ensured that robot performs as it required to collect water hyacinth from the surface, weight calibration and indication and returning along with the path identification.

2.5.4 Environmental testing

Tested in a real stream of water to check the functionalities in a real environment. It was required test several times to overcome the practical issues that we identified.

2.5.5 Safety testing

Finally, the water proofing ability and stability was tested while implementing safe operations in water.



Figure 19

3. Discussion

Several challenges had to be faced and overcome during the development of the robot. Several models and prototypes were designed, developed, implemented and tested in an iterative manner to improve and enhance the solution, resolving issues that were identified.



Figure 20

3.1 Structure

At first, structure was developed using polystyrene (EPS) and supported by wooden sticks. But it was not stable or waterproof. Structure was damaged during transportation and was difficult to float on the water surface. Then the structure was recreated using PVC pipes. Pipes were connected using L joints and T joints to make it a firm and stable structure. Fore and rear

compartments were supported by polystyrene (EPS) to make it more lightweight. It provides an additional support to float in water, and it also protects components from the water.

3.2 Water hyacinth collection

Initially, conveyor belt was implemented using a cloth attached to an elastic belt which was not stable and not strong enough to carry plants towards the bin. Tangling nature of the plant also prevented the collection of the plants to the conveyor belt. Then it was replaced by the 22 iron gauges which is firmer and more stable while attaching sharp ends on wires to grab plants from the surface. During environmental testing belt, we identified it needs further support for the collection. Then bucket lift was initiated as a further support for the process to operate with the help of the servo motor.



Figure 21

Additionally, gear motor was replaced by the gear motor -12V 200RPM XC25GA370, since it required more power to rotate the conveyor while collecting plants.

3.3 Weight calibration and indication



Figure 22

Middle compartment was created using a net at the beginning. But for the weight calibration it was required a stable surface. Then the net was replaced with a plastic surface along with the wooden plank by placing the load sensor underneath and the collecting bin on the top. It provided an accurate reading of the collecting plants weight and indicate with red bulb and the buzzer when maximum weight reached.

3.4 Path identification

To make it a self-driving robot several functionalities were implemented. To identify the edges or the ends of the water body, push buttons were selected to place on the front and both sides of the robot after testing several components. When buttons pressed against the hard edges it has been commanded correct its path using the coding.

Returning process is also automated as it reached the maximum weight robot is returning to the initial position by evaluating the current position and the time spent for the journey from the starting position. But it is not an accurate method for the process.

3.5 Power supply.

Initially components were powered by the microcontroller. Due to lack of power and regulation issues, it is supplied using external batteries for the motors. Since the power supply was not enough to rotate the relay module was used. To protect components from high voltages and to supply regulated power, buck converters were implemented for the LCD panel and the load cell. When connecting some components jumper wires were replaced by the circuit wires.

3.6 Control unit

Rear compartment consists of the control unit including the key components and the power supply. To stabilize the unit plastic containers were used along with the Rexene cloth and disposable polythene bags to cover up the components and protect from water splashes.

4. Future Implementations

1. Ultrasonic sensor detects all the plants and objects on the water body. Implementing a raspberry pi camera module with trained dataset will be able to identify only water hyacinth and collect them.
2. Removal of water hyacinth from the bin is involved with human labour. Removal process can be made it fully automated.
3. Two motors can be installed in front to perform as a cutter to remove heavy plants as a support to the conveyor belt.
4. Implementing the GPS module to locate the robot and give instructions considering the location for the automated removal process.
5. Solar power can be implemented to supply power to the components to avoid the usage of batteries.

5. Conclusion

‘AquaHyel’ successfully collects water hyacinth in the water bodies collect to its bin until it reached the maximum weight limit and then return its original position. And it avoids edges of the water body and correct it path until it finds water plants by the ultrasonic sensor and collect them. Prototype structure has created using PVC pipes and reusable materials as it becomes a cost effective and environmental-friendly solution for the water hyacinth removal. It has some limitations like not using a single power source and using several batteries for components. It also needs a sharp blade or cutter to remove bigger water hyacinth pants. Those limitations can be avoided by further testing and enhancing the robot. Finally, ‘AquaHyel’ robot can be presented as an automated, cost effective, efficient and environmental-friendly solution for the removal of water hyacinth in water bodies.

6. References

1. World Journal of Advanced Engineering Technology and Sciences (2024) ‘Sustainable solutions for water hyacinth invasion: characteristics, impacts, control, and utilization’, *World Journal of Advanced Engineering Technology and Sciences*, 12(1), pp. 47–58. doi: 10.30574/wjaets.2024.12.1.0169.
2. Kumar, G.D., Sreedeharan, B.A., Kiran, T.S., Sandeep, S.S. and Mouli, S. (2020) ‘Design and fabrication of an automatic water hyacinth removal and prevention machine’, *International Journal of Innovative Technology and Exploring Engineering*, 9(5).
3. International Research Journal of Engineering and Technology (IRJET) (2025) ‘Arduino-controlled boat for floating waste collection in rivers’, *International Research Journal of Engineering and Technology (IRJET)*, 12(6), p. 1104.

7. Gantt chart

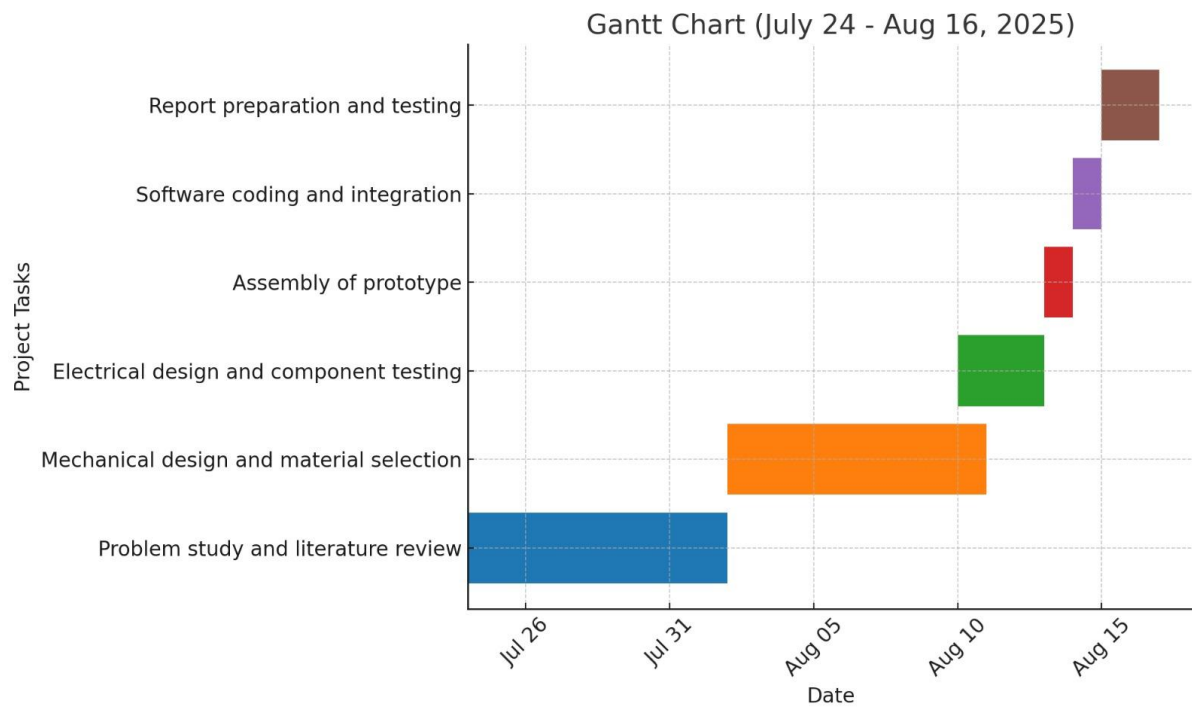


Figure 23

8. Appendix

Video Demonstration Link –

https://drive.google.com/drive/folders/1vtCM02o8gL-2jclU-FXJSbPxeh14Gi_C?usp=sharing